

7. TRAIN AN SSD NETWORK IN A SELF-DRIVING CAR APPLICATION

EX.N0 : 7	TRAIN AN SSD NETWORK IN A SELF-DRIVING CAR APPLICATION
<u>DATE : 11/03/2025</u>	

AIM:

To train an SSD (Single Shot Multibox Detector) model to detect road objects for self-driving car applications.

ALGORITHM:

- Step 1: Import required libraries and modules for SSD and image pre-processing.
- Step 2: Load and pre-process dataset (e.g., Pascal VOC or a custom self-driving dataset)
- Step 3: Define or load a pre-trained SSD model (e.g., SSD300 or SSD512).
- Step 4: Configure loss function and optimizer for training.
- Step 5: Train the SSD model with bounding box labels and object classes.
- Step 6: Evaluate and visualize model predictions on test images or video

PROGRAM:

```
import cv2
import numpy as np
import tensorflow as tf
MODEL_PATH = "C:/Users/AI-LAB/Documents/saved_model/"
detect_fn = tf.saved_model.load(MODEL_PATH)
TARGET_CLASSES = {1: "Person", 3: "Car", 4: "Motorcycle", 6: "Bus", 8: "Truck"}
def preprocess_image(image):
    """Preprocess image for SSD input"""
    image_resized = cv2.resize(image, (300, 300)) # Resize for SSD
    input_tensor = tf.convert_to_tensor(image_resized, dtype=tf.uint8)
    input_tensor = input_tensor[tf.newaxis, ...]
```

```

return input_tensor

def detect_objects(image):
    """Run object detection and filter relevant classes"""
    input_tensor = preprocess_image(image)
    detections = detect_fn.signatures["serving_default"](input_tensor)
    boxes = detections["detection_boxes"].numpy()[0]
    scores = detections["detection_scores"].numpy()[0]
    classes = detections["detection_classes"].numpy()[0].astype(int)
    filtered_boxes, filtered_scores, filtered_classes = [], [], []
    for i in range(len(scores)):
        if scores[i] > 0.5 and classes[i] in TARGET_CLASSES: # Confidence threshold
            filtered_boxes.append(boxes[i])
            filtered_scores.append(scores[i])
            filtered_classes.append(classes[i])
    return filtered_boxes, filtered_scores, filtered_classes

def draw_detections(image, boxes, scores, classes):
    """Draw bounding boxes on detected objects"""
    height, width, _ = image.shape
    for i in range(len(scores)):
        box = boxes[i] * [height, width, height, width]
        y_min, x_min, y_max, x_max = box.astype(int)
        label = TARGET_CLASSES[classes[i]]
        cv2.rectangle(image, (x_min, y_min), (x_max, y_max), (0, 255, 0), 2)
        cv2.putText(image, f'{label}: {scores[i]:.2f}', (x_min, y_min - 10),
            cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0, 255, 0), 2)
    return image

cap = cv2.VideoCapture("C:/Users/AI-LAB/Downloads/Cars_On_Highway.mp4") # Replace
with 0 for live camera feed
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:

```

```
break
boxes, scores, classes = detect_objects(frame)
output_frame = draw_detections(frame, boxes, scores, classes)
cv2.imshow("SSD Vehicle & Pedestrian Detection", output_frame)
if cv2.waitKey(1) & 0xFF == ord('q'):
    break
cap.release()
cv2.destroyAllWindows()
```

OUTPUT:



RESULT:

Thus the Program has been executed successfully and verified.